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Reporting the value of biological findings given activity level propositions: A discussion of published recommendations

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Abstract

Evaluation of DNA given activity propositions (EGALP) is a widely discussed topic at this time, particularly in the United States. There is concern about opinions given in testimony that are not properly founded. Guidance on evaluative reporting given alleged activities can be found in published papers, official documents, and specialized textbooks. In this work, we aim to align and compare recommendations on evaluative reporting of DNA results. Some of these recommendations are explicitly stated while others are woven into the text. All documents pertain to evaluative reporting; they agree on the use of likelihood ratios and the need to avoid the transposed conditional. There is some disagreement such as whether a quantitative or qualitative *LR* should be reported. However, the majority situation is that one topic is covered explicitly by one document but only implicitly covered or not mentioned in the others. We identify 19 consensus recommendations and highlight five gaps or areas of disagreement for which we offer suggestions. It is our hope that this will encourage conversations that will lead to a more uniform set of guidelines, perhaps during a periodic updating of existing documents.

KEYWORDS
evaluation of DNA given activity level propositions, evaluative reporting, expert knowledge, forensic DNA analysis, gap analysis, summary of best practices, testimony

Highlights

- We consider evaluation given activity level propositions (EGALP) of biological evidence.
- EGALP for DNA evidence at court is often of interest, indicating the need for reporting.
- Guidance documents pertaining to EGALP are compared.
- Here, 19 consensus recommendations and five suggestions are presented for discussion.
- Efforts to unify existing guidelines may advance the implementation of EGALP for DNA evidence.

1 | INTRODUCTION

The evaluation of biological findings given activity level propositions (hereafter EGALP) refers to a comparison of the probabilities of some observations (commonly referred to as results, findings, or the evidence) given at least two mutually exclusive propositions regarding disputed activities that might have resulted in the evidence. The authors are aware that this has been commonly referred to in some writings as "Activity Level Interpretations" or "Activity Level Reporting." It is often misunderstood that by performing these

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evaluations we are able to discuss the activities. Therefore, this shorthand will be avoided in order to emphasize the necessity of focusing on the value of the results in light of the activity level propositions – not the activities themselves.

Recently a report by the Texas Forensic Science Commission [1] discussed a complaint that alleges:

> ...activity level analysis and testimony departed from "established best practices," including:
>
> - preparing a written report;
> - researching relevant literature;
> - assigning probabilities based on the literature;
> - delineating the specific studies relied upon in formulating an opinion; and
> - selecting balanced propositions.

This report has been influential in the US placing a considerable barrier to any consideration of EGALP. Any consideration of this complaint and subsequent report requires some definition of established field best practice.

Several guidance documents relating to reporting evaluation given alleged activities have been published. In addition, some documents of a more general nature have sections on reporting evaluation given alleged activities. These include:

1. The report of the International Society of Forensic Genetics' (ISFG) DNA Commission [2] (hereafter ISFG guidelines),
2. The European Network of Forensic Science Institutes' (ENFSI) guidelines for evaluative reporting [3] (hereafter the ENFSI guidelines),
3. Best Practice Manual for Human Forensic Biology and DNA Profiling ENFSI-DNA-BPM-03 Version 01 – December 2022 (hereafter ENSFI-BPM) [4],
4. The Forensic Science Regulator (United Kingdom) Codes of Practice and Conduct Development of Evaluative Opinions [5] (hereafter FSR),
5. The Report of the Expert Working Group on Human Factors in Forensic DNA Interpretation [6] (hereafter the NHF report),
6. OSAC 2022-S-0024 Best Practice Recommendations for Evaluative Forensic DNA Testimony Draft (hereafter OSAC) [7], OSAC authorship available here: https://www.nist.gov/osac/subcommittees/human-forensic-biology.
7. National Institute of Forensic Science Australia New Zealand, An introductory guide to evaluative reporting [8] (hereafter NIFS),
8. NIST Foundational Review [9] (hereafter NFR),
9. Communicating and Interpreting Statistical Evidence in the Administration of Criminal Justice 4. Case Assessment and Interpretation of Expert Evidence Guidance for Judges, Lawyers, Forensic Scientists and Expert Witnesses Graham Jackson, Colin Aitken and Paul Robert (hereafter RSS), [10]
10. Standards Human DNA Analysis and Interpretation. Regulation (hereafter NRGD) [11].

Additionally, there is a large body of published material that is of great value for any consideration EGALP. This would include, but certainly is not restricted to, Taylor and Kokshoorn [12] (hereafter T&K), Aitken and Taroni [13], and the early papers setting up the framework [14–16].

The ISFG document is written specifically for EGALP. Most of the other documents are broader in scope. FSR, ENFSI, ENSFI-BPM, NRGD, NHF, OSAC, NIST, RSS and NIFS all have sections dedicated to EGALP. The UK regulator and ENFSI have consulted widely on the creation of the guidelines. The NFR underwent a public comment period followed by extensive revision, and the NHF had "a panel of independent experts [which] supplied written comments on a completed draft" (see page 2 of [6]), but did not undergo either a public comment period nor a traditional blind peer review. The draft OSAC document was open for a public comment and revision process and underwent a Scientific Technical Review Panel review. At time of writing, the OSAC document is under revision.

We have reflected on whether the focus of the documents on general evidence evaluation or specifically on EGALP should affect the recommendations and we think that it should not.

There is some alignment between the documents but also some key differences. Given the variations, it would be difficult to say exactly what the recommended best practice is, although themes certainly exist. We do not suggest that any one of these documents, or the one we are writing here, should assume the title of best practice. Science constantly evolves, and progress is made through criticism, reflection, experience, research, and further publications.

In this paper, we attempt to discuss and compare these documents with additional input from the general literature. We attempt to hold back our own opinionizing and to try to be transparent in the few times this occurs.

We have taken the clearly marked recommendations from these documents and attempted to align them into topic areas. In most cases, an explicit recommendation exists in one or a few documents but not in the others. In these cases, we have searched for text in these documents with equivalent meaning. We have grouped both the explicit and the implicit recommendations into topic areas and distilled from them a consensus recommendation. We have attempted to order them in a sensible manner; however, we recognize that not all readers will agree with this organization. In some circumstances, we detect topic areas that are not covered in depth or even at all by these documents. In such cases, we have proposed suggestions to promote further discussion in these areas.

We do not repeat the full content of the section of each document applicable to each topic, as the relevant sections of these documents total well over 300 pages, not counting T&K, which is over 500 pages itself. In most cases, we select one or a few and reference the relevant section of any other documents that discuss this topic. A glossary of terms used in this paper is provided in Table 1.

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TABLE 1 Glossary.

ALQ	A term coined in this paper for activity level Questioning during testimony
EGALP	A term coined in this paper for Evaluation of DNA Given Activity (Level) Propositions. This is more commonly referred to in the literature as activity level reporting (ALR)
ENFSI	ENFSI are the European Network of Forensic Science Institutes. In this paper this refers to their guidelines for evaluative reporting [3]
ENFSI-BPM	ENFSI Best Practice Manual for Human Forensic Biology and DNA Profiling ENFSI-DNA-BPM-03 Version 01 – December 2022 [4]
Evaluative opinions	A term used to describe opinions for court use on the weight of evidence
FSR	In this paper this refers to the Forensic Science Regulator Codes of Practice and Conduct Development of Evaluative Opinions [5]
FSS	The Forensic Science Service (FSS) was a government-owned company in the United Kingdom, which provided forensic science services to the police forces and government agencies of England and Wales, as well as other countries. It closed in 2012
ISFG	In this paper this refers to the report of the DNA commission of the International Society for Forensic Genetics Part II [2]
NFR	NIST Foundational Review [9]
NIFS	The National Institute of Forensic Sciences, Australia and New Zealand [8]
NIST	The National Institute of Science and Technology
NFI	Netherlands Forensic Institute
NHF	The report of the NIST Human Factors working group [6]
NRGD	Netherlands Register of Court Experts Summary of concepts Human DNA Analysis and Interpretation [11]
OSAC	Best Practice Recommendations for Evaluative Forensic DNA Testimony Draft [7]
POI	Person of interest
RSS	Communicating and Interpreting Statistical Evidence in the Administration of Criminal Justice, 4. Case Assessment and Interpretation of Expert Evidence, Guidance for Judges, Lawyers, Forensic Scientists and Expert Witnesses, Royal Statistical Society [10]
T&K	Taylor and Kokshoorn [12]
TPPR	Transfer, persistence, prevalence, and recovery of biological traces

2 | COMPARISON AND CATEGORIZATION OF RECOMMENDATIONS

2.1 | Principles

ISFG [2], FSR [5], and ENSFI [4] all advise applying the principles of interpretation given by Evett and Weir. We give these principles as Consensus recommendation 1. Principle 2, as phrased, is modified or expanded on in many of the documents, for example, advising what to do if the defense position is unknown or advising that, even if Hd is known, if the scientists can think of a better Ha, then that should be used.

Consensus recommendation 1: Principles of evidence evaluation:

1. Interpretation takes place in a framework of circumstances.
2. We must evaluate our results given two mutually exclusive propositions representing the views of the two involved parties as understood.
3. Scientists must give their opinion on the *results* and not on the propositions.

2.2 | Quality assurance/quality control

2.2.1 | Training, expertise, and competency

Four documents FSR, ISFG, NHF, and ENFSI do mention education and expertise, but do not specifically outline expectations for training. Taylor and Kokshoorn [12] outline some specific expectations and van Oorschot et al. [17, 18] also discuss these needs. NGRD discusses some requirements for minimal numbers of cases to be evaluated each year to maintain placement on the registry but does not address training or education other than to require a master's degree.

There are two areas where training is required:

1. Recognizing when testimony transitions to questions regarding DNA transfer (better recognized as activity level questioning or ALQ) and managing the responses within the demonstrated competencies of the witness.
2. Training to the standard required to do formal EGALP.

According to T&K, a successful laboratory-training module could be:

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11.5.4 Training staff to become activity level evaluators

11.5.4.1 Objectives – At the completion of this unit, the trainee will be able to

- Understand and apply the hierarchy of propositions.
- Identify cases where activity level propositions will have a significant impact of evidentiary evaluations.
- Understand the evaluation of evidence using Bayesian reasoning, including the formation of reasonable alternate activity propositions [].
- Demonstrate an ability to create Bayesian networks using [] software for the evaluation of evidence given specific case circumstances.
- Demonstrate an understanding of the types of data required in the various components of the evaluation and the ability to locate that data either from literature or in-house sources.

NHF states: 1.3 Forensic practitioners will not report on matters outside their own area of expertise.

FSR sections 6.1.6 and 8.1.1 also explicitly state that witnesses must opine only on areas in which they have been shown to be demonstrably competent.

The Netherlands has an expert witness register (NRDG, https://english.nrgd.nl/) with a category for "DNA Activity Level." To be registered as an expert requires proof of competency through provision of documented evidence of training to complete evaluations, accreditation (if available), organizational authority to report at activity level (if available) and recent cases. All of these documents are assessed by independent assessors registered with the NRGD.

It will be necessary to define the competencies and the tests that demonstrate competence. This is likely to already be done in accredited labs but would need extension to accommodate independent scientists or those working in non-accredited laboratories.

Consensus recommendation 2: Experts shall only answer questions that they have been shown to be qualified to answer and shall be able to recognize the progression from simple answerable questions to ones that require additional work, data, or training. Experts undertaking formal EGALP shall be trained to do so. They shall have demonstrated competency.

2.2.2 | Proficiency tests

Not all of the recommendations address proficiency testing. ISFG, ENFSI, and NIFS mention the need for proficient or competent personnel; OSAC is silent (although that document specifically deals with communication at court). NFR has extensive discussion of proficiency testing for sub-source propositions, but is silent in Chapter 5, which covers activity level. The NHF report states that proficiency testing should be "ongoing", leaving FSR as the only recommendation to directly address the issue.

FSR: 8.1.2 b. Experts should participate in regular calibration of their expertise through, for example, blind proficiency tests that are representative of the complexity encountered in casework.

We do not think that this represents a current consensus for proficiency tests for EGALP, and we are not aware of any provider of such tests.

2.2.3 | Peer Review

Peer review is the norm in accredited laboratories and is undertaken by an examiner qualified in the subject matter. It is not the norm, however desirable, for independent scientists.

NHF does have a section on technical peer review. This section refers to DNA evidence in general and not specifically to EGALP. They do state the following with regard to testimony that is given without a formal EGALP: "Moreover, it would be problematic to give an opinion on the value of their results considering factors such as transfer and background, without having sufficient validated methods, training and education, competency testing, and quality system assurances available." This comment does suggest a need for peer review and proficiency testing.

ENFSI have a section on case file notes stating that the case file "is peer reviewed by another expert."

T&K 11.5.2 give a list of expectations for the content of a peer review (see also FSR 8.1.1g and NHF Recommendation 7.2, which support the need for peer review but do not give explicit lists). The T&K list includes (edited for brevity):

- Copies of referenced literature present?
- Conclusions and expressions justified and correct?
- Limitations and assumptions stated? All literature used referenced?
- Relevant data used to populate probability tables?
- Correct information entered into [Bayes] network?

NIFS gives: "Policies, procedures and reporting standards should be prepared to support consistent application of evaluative reporting methodology among practitioners. This should include procedures for peer review and resolution."

These documents appear to be discussing the situation where evidence is prepared well ahead of time in an accredited laboratory and a report is issued, although this is not stated.

Consensus recommendation 3: All reports shall be peer reviewed.

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We separately discuss the situation for cross-examination and rebuttal.

2.3 | Case management

2.3.1 | Pre-assessment

We could not find a definition of case pre-assessment in the documents examined so we provide our own.

Case pre-assessment involves determining the questions to be answered, the possible findings, and assigning the probability of these findings before the findings are known. Its purpose is to ameliorate bias in any subjective assessments and to guide the scientist and stakeholders as to whether an analysis is even warranted.

An example is given in ENSFI-BPM which we reproduce in Table 2:

Case pre-assessment is advised in ENSFI, ISFG, and ENSFI-BPM. FSR does discuss it as well but without a clear recommendation to undertake it.

> ENSFI: Case pre-assessment … should be conducted in cases when activity level propositions are set.

> FSR: 2.1.6 The assessment (see 9.1.4) addresses potential outcomes under the conditions of the prosecution and the defense propositions being true. As a result, the expert can advise the commissioning party whether scientific analysis can be of assistance in answering questions of relevance to the case and, if so, enable the appropriate tests to be carried out.

> ISFG: In a given case, because of post-hoc rationalization, it will be important that one assigns the probability of recovering say "a major profile" before knowing what the actual results are.

> ENSFI-BPM: Pre-assessment can be used (as a means) to specify the main potential findings from examinations of the items submitted and then to assign their probabilities considering each proposition.

We summarize these in Consensus recommendation 4.

Consensus recommendation 4: Case pre-assessment should be conducted in cases when activity level propositions are set.

2.3.2 | Proceed/do not proceed

Three of the selected documents have explicit guidance on a proceed/do not proceed decision.

FSR 8.2.6, ENFSI 1.4, and T&K 11.5.1 and 11.5.3.5.2 have explicit advice regarding a proceed/do not proceed decision.

> FSR: 8.2.6 The expert shall consider, firstly, whether the relevant case circumstances allow them to specify the fact in issue … at the activity level and, secondly, whether they then have the expertise and resources to provide an opinion to help address that fact in issue.

This paragraph is our own writing synthesized from the various documents. Whether or not an evaluation of the observations given the competing activity level propositions is of value to the case should not be decided based on the magnitude of the *LR*, or the proposition with which its support lies. There are many cases where the DNA findings will be relatively neutral with respect to the propositions. Often in these cases, the report is most important to provide, lest the DNA profiling evaluation alone be misunderstood by the jury to be highly supportive of an activity [19]. With this in mind, there is a role to play for experts when triaging cases to guide which exhibits are examined so that those with very little power to discriminate between activity level propositions are not accepted in the first place.

An example of expert evaluation in determining whether to proceed is discussed by ENFSI: *3.4 If, as a result of the pre-assessment, scientific examinations are unlikely to assist in differentiating between the propositions, the mandating authority or party will be advised accordingly. It is recommended that such advice and the result of it are documented in the case file*.

Consensus recommendation 5: The expert shall decide whether or not to proceed with EGALP based on the availability of expertise and resource, and on the potential to deliver value.

TABLE 2 Case example of pre-assessment where DNA is recovered from the fingers of the accused.

Outcomes (E)	Pr(E|Hp,I) if digital penetration	Pr(E|Hd,I) if social activities	Likelihood ratio Pr(E|Hp,I)/Pr(E|Hd,I)
Large quantity & full female profile	0.82	0.16	≈ 5
Small quantity & partial female profile	0.09	0.16	≈1
No female profile	0.09	0.68	≈1/7
Total	1	1	

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2.3.3 | Base the propositions on the case circumstances and/or instructions from the prosecution or defense

Again, some of the documents give explicit guidance of the use of the case circumstances and/or instructions from the defense to form the propositions.

The FSR guidance is explicit stating:

> 8.2.12 The expert shall set out clearly what propositions have been addressed so that the defense, or the court, can determine if the report addresses issues of relevance to the case.

The ENFSI state:

> 3.12: Reports should include (not exhaustive list):
>
> - Conditioning information used
> - Mandate and questions asked, if required
> - The propositions of interest

OSAC gives:

> 4.4.2.1.1 Propositions should be based on case information and the issue where forensic science can help, not on the biological results.

See also T&K pg. 17 and all of chapter 4.

The FSR guidelines agree:

> 8.2.9: When no defence proposition has been offered, the expert should generally adopt, on behalf of the defendant, one or more alternative propositions, which should be relevant to the facts in issue. The propositions shall be clearly stated in the report. An alternative proposition may be a form of negation of the prosecution proposition, which may be qualified (e.g., the DNA is not from the suspect but from an unknown person) and may take into account the case circumstances, i.e. the defendant (who was arrested at his home) has had no involvement in the events. The limitation of this approach is that a simple negation of the prosecution proposition may tend to maximise the value of the LR. As a result, this limitation should be stated clearly within the expert's report, with an explicit offer to re-evaluate should an alternative proposition be provided.

OSAC again gives:

> 4.4.2.1.5.2 When no specific alternative is available from the defense, the expert should adopt a reasonable proxy proposition based on the case information in order to fully evaluate the evidence.

The following paragraph is our own writing synthesized from the various documents. However, we recognize that, in any adversarial system, the defense has no requirement to disclose their proposition. If an evaluation of the results given activity level propositions is being undertaken without disclosure by the defense, as is their right, then a sensible proxy proposition that safeguards the legitimate interests of the defense should be used. As an example, in South Australia, there is a policy that the defense is contacted at the start of the reporting process and is invited to give input. They may or may not respond to this invitation, but it is offered nonetheless. Experience in working in forensic DNA laboratories in the USA suggests that this may be difficult to implement at a similar stage in the process, as often no defense attorney is assigned to the case until well after the case has cleared the forensic laboratory. However, some policy stating that the defense can provide input before or after the report is issued is reasonable.

Consensus recommendation 6: The propositions should be based on case information and formulated according to published guidelines; for example, they shall not include explanations of how the DNA transferred or results obtained.

Most of the documents state that a sentence shall be included in the statement that if the information changes, then the interpretation will need to be reconsidered.

> FSR 8.5.6: There shall be, in the report, a sentence saying that if the information (particularly that on which propositions are based) changes, then the interpretation will need to be reconsidered.
>
> ISFG: A caveat in the statement will reinforce the point that evaluation is strictly conditioned upon the case information that has been put forward.
>
> ENSFI: It (conditioning information) shall always be regarded as provisional, and the examiner shall be ready to re-evaluate findings if the conditioning information changes.
>
> ENSFI-BPM: A caveat will in this case indicate that if the information changes, a new evaluation will be needed.

Consensus recommendation 7: There shall be, in the report, a sentence saying that if the information (particularly that on which propositions are based) changes, then the interpretation will need to be reconsidered.

Several of the documents discuss that the report should outline the limitations of any evaluation that is restricted to sub-source level propositions. When reporting in a manner restricted to sub-source level propositions, in addition to avoiding transposition errors, care should

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be taken to avoid any suggestion that evaluation considering activity level propositions *cannot be done* as opposed to *has not been done*. Also, any implication that any competing explanations are *equally likely to have occurred* (which is a transposition) or that the evidence is *equally likely under each explanation* (which is not addressed in sub-source reporting) should be avoided since either option would result in misunderstanding by the fact finder. We quote ISFG exactly in the consensus recommendation, but similar material also appears in FSR and NHF.

Consensus recommendation 8: To make clear the limitations of providing a report that is restricted sub-source level, the following caveat can be applied to statements:

> This report does not provide any information on the mechanisms or actions that led to the deposition of the biological material concerned. It only provides help regarding its origin. Should there be any issue regarding the activities that led to this material, we should be contacted to provide a new report.

2.3.4 | Discuss/meet with the defense

> FSR: 8.2.7 If possible, and in the interest of a balanced approach, the expert shall seek the other party's version of events from the investigator, prosecution team, or defense team (see section 6.2).
>
> 8.2.2 The expert shall: …

b. Consider all available, relevant case-specific information and, where necessary, request additional information; and
c. Discuss the issues to be addressed and potential propositions with the relevant instructing party (e.g., police, defense, prosecuting authority) and, where possible, the other party.

> T&K: 11.5.3.6 Setting activity level Propositions.
>
> 11.5.3.6.1 Communicating with prosecution and defense
>
> <snip>
>
> Much of this information will be available on the police submission forms and/or medical examination notes however further discussion with stakeholders will be required for transparency and to encourage further input. The two mutually exclusive propositions that will be considered should be provided to opposing parties to ensure their standpoints are captured accurately. If the work is being requested by the police, prosecution or defence, it is critical to seek approval from the requesting agency to disclose the propositions with the other party.

The following paragraph is our own writing. Among the authors of this document, those of us who testify in the US have never contacted the other party directly. We assume that any defense contact would need to go via the prosecutor. On rare occasion, the other counsel may choose to contact the expert directly. From a practical point of view, there is often no mechanism for the expert to identify the defense counsel, and there may be significant delays in the court systems, which contribute to the challenge of contacting the defense. We know of some laboratories that have policies that allow communication with the defense but this is not a universal policy. Even when the defense contacts the expert ahead of time, based on our experience, often times there is little information communicated by the defense—as is their right.

Consensus recommendation 9: The defense should be given an opportunity to comment as early as realistic on the case information presented as understood, propositions, data sources, and any assumptions made.

2.4 | Setting the propositions

Many of the documents mentioned do discuss the propositions, often stating such things as the level in the hierarchy to address, what to do if the defense offers no proposition, and advice not to change the propositions after observing the findings. Only two of the documents [2, 12] give explicit advice on forming the propositions. We follow T&K here. They have an entire chapter on this subject (Chapter 4) which we summarize down to essentially the section headings. Since this subject is not discussed extensively in the other documents with the exception of ISFG, it is difficult to describe it as a consensus, although we suspect that it is. We label it here Suggestion 1.

Consensus recommendation 5 advises giving the defense an opportunity to comment as early as realistic on the case information presented as understood, propositions, data sources and any assumptions made. T&K make the valid observation that "Based on the context of a case, the scientist must consider all relevant routes for TPR of biological traces. It is therefore crucial that, if such routes are overlooked by the parties involved, the scientist makes these explicit."

Suggestion 1: Propositions must be:

1. Relevant to the court.
2. Consider all relevant routes for transfer, persistence, and recovery of biological traces.
3. Must be specific enough that assigning a probability of the evidence given the truth of them is possible.
4. Come in (at least) pairs, (the "at least" is added by the authors of this document).
5. They must be exclusive and should be exhaustive within the context of the case.
6. They should not include results or mechanisms.
7. They should be concise.

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2.5 | Assignment of the LR

2.5.1 | Use a likelihood ratio

All documents recommend the use of an *LR* (ISFG recommendation 2, FSR 8.3.1 e, ENFSI 3.1.4, ENSFI-BLM pg. 26, NHF recommendation 4.1, NRGD section 2.4, T&K 11.5.3.6.4). NHF: *Recommendation 4.1: Forensic science service providers should use likelihood ratios to evaluate DNA results*.

Consensus recommendation 10: A likelihood ratio shall be assigned which compares the probability of the findings given the first proposition (and considering the available case information) to the probability of the findings given an alternate proposition (still considering the available case information). The propositions are ascribed to the prosecution and alternative (ideally defense) positions, respectively. The *LR* is conditional on both the propositions and case information.

2.5.2 | Contamination

Contamination is mentioned by ISFG section 4 pg. 3 "Are there other issues, which the scientist should consider, such as the possibility of contamination?", T&K pg. 43, NHF callout box 7.2 and ENFSI 3.7. However, there is no clear recommendation what, if anything, should be done, and whether the possibility of contamination should be included in the construction of an *LR*.

Suggestion 2: If the analysis assumes that no contamination has occurred, then this shall be explicitly written into the notes and the report.

2.5.3 | Undertake a sensitivity analysis

A sensitivity analysis varies the probabilities assigned to certain variables across the plausible range and notes the sensitivity of the *LR* to various factors:

> ENSFI-BPM: a sensitivity analysis may be carried out to show the effect upon the LR.
>
> ENSFI 3.10: According to their uncertainty, forensic practitioners should consider exploring the sensitivity of the likelihood ratio to different probabilities by examining the effect of assigning different probabilities.
>
> ISFG: Taylor et al. [59] point out that the effect on the value of the results (LR) may be explored with sensitivity analysis as described in the supplement, fig. 3. Sensitivity analysis [63] offers a transparent approach because the effect of different factors on the evaluation can be simulated.

T&K extensively discussed in their Chapter 8:

> Ultimately the depth and breadth of sensitivity analysis carried out are similar to any other aspect of evaluation (i.e. the resolution of the data used, the assumptions made, the simplifications, the breadth of different transfer mechanisms). It will depend on the case and the assessment of the analyst as to what is required. It is likely that, at a minimum, a basic test of different probability values will be carried out whenever data is sparse. Questions that have arisen from prosecution or defence may also guide the analyst on the aspects which are likely to be explored in greater depth. Whatever testing is carried out beyond this is a personal and professional choice.
>
> NHF: Because probabilities (and LRs) depend on knowledge, data, and modeling assumptions, sensitivity analyses can be used to evaluate the impact that changes to these dependencies will have on the LR value.

Sensitivity analysis is not mentioned in FSR. NIST mentions that there is research into sensitivity analyses.

Suggestion 3: A sensitivity analysis may be performed. If performed, it should be recorded in the notes and signaled in the report.

2.5.4 | Source of the data

Most of the selected documents discuss potential sources for the data. For example: ISFG 7.1, FSR 8.3.11–13, ENFSI guidance note 3, and T&K 11.5.3.6.5.

> FSR: 8.3.10 Published data of appropriate quality should be used wherever possible as a basis for these assignments. If relevant published data are not available, or to supplement published data, data from unpublished sources may be used as long as the provenance and quality of the data have been reviewed and deemed suitable by the expert and the source of the data is documented on file. Unpublished datasets shall be made available for inspection by an expert instructed by the other party if probability assignments based thereon are a fact in issue in the case.

When no appropriate published data are available, IFSG (7.1) allow for "calibrated experience" and "peer consultation" and ENFSI (Guidance Note 3) allows for "personal data such as experience in similar cases and peer consultations" to be used.

Data relevance and quality are also discussed:

> UK regulator: 7.2.7 In an ideal situation, the scientist would have access to relevant, high quality data sets

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to inform their evaluation. However, the nature of forensic science, where every case has different circumstances, means that large and directly relevant data sets are very rarely available. A LR determined by the expert using experience alone can be no more wrong than the expert's opinion on which it is based, and that opinion is admissible evidence. However, it is less easily tested than a LR based on relevant data, quantitative measurements, and statistical models.

T&K state pg. 172: There is no specific cut-off or delineation at which a study becomes too different to the case circumstances in order to use. It is often the case that some probabilities will simply have no direct data that can inform the probability assignment. In these cases, the best the analyst can do is finding one or more pieces of information from less directly applicable studies that will assist in the assignment.

Consensus recommendation 11: Sources of data should ideally be a study that has conditions that match those of the case. Failing the availability of such data, the scientist may use secondary sources or expert opinion. Regardless of the source, it should be transparent within the report how the probabilities have been assigned.

2.5.5 | Assess all the findings

All the findings should be considered where practical (See for example ISFG recommendation 10, ISFG consideration 3 pg. 10, T&K pg. 46). If evidence is absent which we would expect to be present under H_p, then this absence of evidence supports the defense proposition. Even if a weight cannot be applied by this witness to the absence of this evidence, the court should be informed that it is exculpatory, or that another witness may be required who can give the testimony that this evidence is exculpatory.

Consensus recommendation 12: All results (e.g., presence and absence of DNA) shall be included in the observations being evaluated, or their omission (for example due to the irrelevance to the disputed activity) should be explained and justified.

2.5.6 | Contents of the case file

There is some guidance on the contents expected for the casefile. We note that the general expectation for the content of the casefile should be broadly consistent across evidence types.

ENFSI: 3.11 The case file should include (not exhaustive list):

1. Case information (verbatim, or as otherwise received)
2. Mandate and questions asked, if available
3. Materials and items received
4. The key issue(s) and propositions of interest
5. All discussions with mandating authorities and parties in the case
6. Examination strategy
7. Methods used
8. Potential outcomes and assigned probabilities at the time when pre-assessment was carried out
9. Relevant data used in probability assignments [guidance note 3]
10. Observations made and analytical results
11. Discussion and evaluation of the strength of support that the findings provide to help to resolve the issues (and related propositions) dictated by the purpose and the circumstances of the case
12. Conclusions and report given to the mandating authority or party.

Item number 8 in the list above (Potential outcomes and assigned probabilities at the time when pre-assessment was carried out [15]) refers to an activity that is not commonly undertaken by the examining scientist in the USA. Pre-assessment was encouraged via the Case Assessment and Interpretation approach adopted by the FSS in the late 1990s. [15] This process of recording a set of expected observations *prior to the examination* and then assigning probabilities to those outcomes allows an expert to later assign an *LR* that has not been influenced by the observations. This can be a useful tool but is not typically part of the examination process in the US in terms of some specific evidence item in the context of case information that might yield an *LR* for EGALP.

Instead, pre-assessment in the US is most used when making case management decisions related to which evidence items to examine (as opposed to returning to the submitter with no examination) or whether to accept the case at all. Occasionally, pre-assessment (and whether to proceed with an examination) is used when the probabilities being assigned are largely or wholly based on expert knowledge rather than on published data or are expected to be very close to 1 or 0 (i.e., testing a vaginal swab in a sexual assault case where victim claims completed intercourse and defense claims no sexual activity occurred). We are aware that in some laboratory systems, case triage and pre-assessment are undertaken by a different individual than the expert who reports the results in an attempt to reduce unconscious bias in the interpretation of evidence. We suggest that the pre-assessor's work should be made available to the reporter during the evaluation stage and, of course, would be disclosable.

Case file expectations also appear in NHF at various places and T&K 11.5.3.5.3.

Consensus recommendation 13: The case notes shall contain, at a minimum, the background case information, the propositions, all of the published material or sources of data, the logic, formulae, or Bayes Net used, the resulting *LR* or the results of the sensitivity analysis.

2.6 | Reporting

2.6.1 | Proper phrasing of the conclusions

This is a rare situation where advice is unanimous. See for example FSR 7.1.1, ISFG recommendation 3, NHF pg. 176, ENFSI pg. 27, ENSFI-BPM pg. 26, T&K pg. 49.

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The ISFG guidelines [2] state:

> Recommendation 3: Scientists must not give their opinion on what is the "most likely way of transfer" (direct or indirect), as this would amount to giving an opinion on the activities and result in an example of what is commonly labeled prosecutor's fallacy (i.e., give the probability of the hypothesis given the evidence, or more commonly the probability that the prosecution hypothesis is true). The scientists' role is to assess the value of the results if each proposition is true in accordance with the likelihood ratio framework (the probability of the results if X is true and if Y is true).

ISFG give an example of a paragraph that could be included in every statement:

> Please note that our laboratory does not provide any assessment on how likely it is that the first proposition or the alternative proposition is true. Indeed, this probability (e.g. The defendant opened the victim's door, i.e., the DNA resulted from primary transfer) is the domain of the court: one needs to combine all the information of the case in order to make such a statement.

> NHF [6]: Recommendation 7.1: DNA analysts should not opine about the possibility or probability of direct or indirect transfer having occurred in a case.

> T&K [12] state: The scientist restricts themselves to expressing an opinion on the weight of the findings given the propositions, and not on the propositions themselves.

> FSR: 8.6.7 The expert shall ensure that they do not illegitimately transpose the conditional and that they make reasonable efforts to correct such a transposition if it is made by any court participant.

> ENFSI Guidance note on logic: Evaluative reports should address the probability of the findings given the propositions and relevant background information and not the probability of the propositions given the findings and background information. The report should not contain statements that are transposing the conditional. (Emphasis in the original).

> OSAC gives: 4.4.2.2 The expert should only make statements about the probability of the evidence given a proposition … The expert must avoid making any statement about the probability of a proposition given the evidence (i.e., transposing the conditional) … Equally, the expert must avoid giving any statement on the probabilities of the propositions themselves.

> 4.4.2.2.1 The expert should be aware that most hypothetical questions are asking for an opinion about the proposition (or explanation), not about the evidence.

> 4.4.2.2.2 The expert must avoid transposing a conditional when answering direct questions from one of the parties as well as when considering both propositions.

Consensus recommendation 14: The phrasing of the probabilistic statements in the report, notes, and testimony shall always be of the form: the probability of the results given that each (in turn) of the propositions is true. The phrasing of the conclusion shall pertain to the ratio of two probabilistic statements regarding the results.

2.6.2 | Written report and contents

Three documents explicitly mention the need to issue a written report.

> ENFSI: 3.1.4 This document is based on there being an appropriate framework in place to guide the handling of a case from initial receipt of a request through to the provision of evidence in court. Stages of that framework include: 1… f. Writing a report; …

> OSAC gives: 4.4.1 Laboratories should establish policies regarding testimony given activity level propositions when this is of interest to the court. It is best to have previously reported any relevant opinions in writing as well; however, guidance on the determination and reporting of these opinions is outside the scope of this document.

> NIFS gives: Forensic practitioners normally prepare a report to assist the triers of fact in their adjudications. The report is an explanation of the whole process, including the information and method used to form propositions, the available case items, the analyses carried out on them, the resulting observations, the method of evaluation and the conclusions drawn from it.

The authors interpret these three documents as a report being mandatory (ENFSI) or highly desirable (OSAC and NIFS). We have held back from giving our own opinion, but it would seem necessary for the community to decide if a report is mandatory or optional for evidence in chief. If the subject of EGALP comes up in:

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1. Cross-examination, or
2. Rebuttal with no time to prepare and has reviewed a report

then it is impractical to issue a report. There is some discussion of cross-examination (see later) but the community could strengthen its recommendations in this area.

Consensus recommendation 15: A report shall be issued wherever practical. The report shall include, at a minimum, the case information relied upon, the propositions used, an assessment of the *LR* in a numerical form. A verbal qualifier is optional. The report shall include all relevant traceability information. The report should invite the defense to have input.

2.6.3 | Traceability and transparency of logic and data

Traceability refers to the ability to track from every statement in the report to the supporting information that underpins this statement in disclosure documents. All the documents summarized here emphasize this aspect.

For example ISFG Recommendation 9:

> In the statement, the source of the data that are used to provide the probabilities should be disclosed, ….

See also ENFSI guidelines pg. 11, ENSFI-BPM pg. 26, FSR 7.1.8, NHF 10.5.4, and T&K multiple places.

> OSAC 4.4.2.3.2 The information relied upon to form the opinion of the expert should be disclosed and clearly communicated.

Champod states [20]:

> Any scientist offering views as to his/her expectations for the forensic findings under given case-related circumstances should be able to put forward documented sets of controlled experiments whose relevancy to the case under dispute can be argued.

We agree, and point out that "relevancy to the case under dispute" is a key component.

For prosecution witnesses there is an expectation of disclosure of all the key elements of anticipated testimony for evidence in chief. For testimony where one needs to account for the possibility of DNA transfer, this would include the propositions, the *LR*, and traceability information identifying the primary data sources, and any subjective adjustments made to assign one's *LR*. There seems to be no consensus that the actual mathematical process is specifically supplied, nor to which degree of detail is necessary should an attempt be made. Dr. Evett reminds us that there is more than just the technical process to consider [21]: "The generation of observations is a technical process and the advances that have been made in forensic science techniques over the last 50 years have been staggering. But science is about reasoning—about making sense from observations." The disclosure should ensure that the scientist's reasoning is transparent and explained.

FSR: 7.1.8 The expert:

a. Provides opinion that includes sufficient information, including limitations, to enable scrutiny by other experts and effective cross-examination;
b. Communicates the reasoning and how they reached their conclusion in a manner that is comprehensible to legal professionals and juries; and
c. Sets out in the report, or any other input into the CJS, the basis of their opinion as described in section 8.5.4.

Consensus recommendation 16: Explicit reference to any scientific publications that give the data used, any unpublished data, any subjective adjustments to the data and the reason for this, and the logic used to assign the *LR* shall appear in the case notes and in the statement, perhaps in an appendix.

FSR also discuss the use of unpublished data: *8.3.10 … data from unpublished sources may be used as long as the provenance and quality of the data have been reviewed and deemed suitable by the expert and the source of the data is documented on file. Unpublished datasets shall be made available for inspection by an expert instructed by the other party if probability assignments based thereon are a fact in issue in the case*. See also ENFSI guidance note 3.

Consensus recommendation 17: Any unpublished data used shall be made available, if necessary under a non-disclosure agreement, for the use of anyone involved in the case. It is acceptable to anonymize DNA profiles. This may be impacted by local laws and ethics considerations.

We have constructed this recommendation from the general content of the documents, but it does not appear explicitly in any of them.

2.6.4 | Quantitative or qualitative LR

The concept that a reported *LR* should be numerical for reporting propositions at the activity level arises, we believe, for the first time in the ENFSI guidelines (2015) [3]. Their clause states:

> 3.14 The conclusion shall be expressed either by a value of the likelihood ratio and/or using a verbal scale related to the value of the likelihood ratio. The verbal equivalents shall express a degree of support for one of the propositions relative to the alternative. The choice of the reported verbal equivalent is based on the likelihood ratio and not the reverse. The report shall contain an indication of the order of magnitude of the likelihood ratio.

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We have placed the and/or in bold face in this clause which we believe suggests that one could assign either a quantitative or a qualitative *LR*. Due to concerns that a numerical *LR* may suggest an evaluation (assigned *LR*) that is "accurate", ENFSI says "in the UK the tendency is to use the verbal equivalent." This means going from a numerical to a verbal *LR*, but not from the verbal to numerical.

A quantitative requirement also appears as Recommendation 11 of the ISFG DNA Commission paper [2]. This recommendation is interpreted as stating that all testimony addressing propositions related to activity level must be numerical and this is confirmed by Taylor (D.A.Taylor, personal communication January 2024) Taylor states that the intent was to facilitate rigor of thinking.

ISFG have a slight internal inconsistency. They state @section 8.2 in relation to the Amanda Knox case:

> Conversely, given that the seized knife was not the murder weapon, if one were to recover blood, it would be present for some unknown reason (i.e. background). … the likelihood ratio must be less than one, supporting the proposition that the seized knife was not the murder weapon. With this example, we see that even if there are no structured data to allow assessment … it is possible to show that the absence of results supports the alternative.

We add that it is important to note that rigor can be employed without the resulting *LR* being numerical and equally a numerical *LR* can be created without rigor.

The FSR (7.2.7), OSAC, NIFS, ENFSI, and NRGD appear to explicitly disagree with ISFG and ENSFI-BPM pg. 26.

> FSR: 7.2.9 Communication of the outcomes of evaluation needs to take account of the fact that lay jurors and other court participants may equate a numerical value of a LR with a level of precision that was not intended by the scientist. For this reason, where the LR is based on a data set that is not adequately representative of the case circumstances or not of sufficient quality or size to enable a precise numerical assessment, it cannot be reported as a single figure; instead, results will be reported at an order of magnitude level, alongside a verbal equivalent…

> FSR: 7.2.10 In cases where the expert has relied solely on observations from expertise that are not available for scrutiny by another expert, quoting any numerical value is not justified and a qualitative evaluation is appropriate…

> OSAC gives: 4.4.3.2 The LR may be communicated numerically (1 million times more likely) or using a qualitative verbal statement (much more likely, somewhat more likely) after assignment of the numerical LR.

> NIFS gives: While this (a numerical LR) is clearly the preferred situation, as it provides the most robust and transparent manner of estimating evidentiary weight, it is possible to estimate the weight of the findings based on experience, published estimates, or combined estimates of probability from multiple practitioners.

> NRGD gives: activity level analysis involves the provision of a numerical and/or verbal strength of support for the probability of the evidence given the truth of two competing propositions formulated at the activity level.

The numerical requirement was novel to some of the authors of this document. Previously, authors of the ISFG guidelines had not stated this. There are arguments in the literature for either approach. For example, Gill states in his 2014 book *Misleading DNA Evidence* [22]:

At Ch 3 @ pg. 77

> The data provided above cannot be directly converted into precise probabilities, but it is reasonable to support broad probabilistic ranges summarized in tabular form: we can potentially assign subjective probability ranges based on symmetric quantiles, where very low is described by Pr < 5%; low describes Pr = 5%–25%; medium describes Pr = 25%–75%; high describes Pr = 75%–95%; very high describes 95%–100%. The wide choice of ranges reflects the uncertainty inherent in the data sets.

Ch 3 @ pg. 78

> Taken in the context of the circumstances of this case and the relative strength of the major component to the DNA mixture, in my opinion the DNA profiling evidence in this case is readily explained and more likely if it had originated as a result of a recent primary transfer event, e.g., scratching between Ms V and Mr S close to the time of her death, rather than as a result of some other type of transfer event which happened 7 days previously or as a result of some unspecified secondary transfer from a biological material deposited in Ms V's house by Mr S on some previous occasion when they were in a relationship with each other.

The NIST foundational review [9] has a chapter on activity level considerations. However, it does not seem to comment specifically either way regarding any requirement for the *LR* to be numerical.

Berger et al. [23] state:

> Nevertheless, it is essential for the expert to address qualitatively their relative magnitudes. Under which of the two propositions is the evidence more probable? That is the proposition which is better supported by the scientific observations. The likelihood ratio is

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> still central, even if the probabilities can only be qualitatively assessed though it is worth reiterating that the expert needs to be able to state the basis for such an assessment.

The position statement of an emeritus group of authors [24] states:

> 5) A verbal scale based on the notion of the likelihood ratio is the most appropriate basis for communication of an evaluative expert opinion to the court. It can be phrased in terms of support for one of a pair of clearly stated propositions.

This ISFG [2] recommendation has been extrapolated to imply that all statements about activity level propositions must be numerical and based on data. We believe that the statement was meant to enforce rigor in thinking and to oppose unfounded speculation. However, it has also been extrapolated to imply that even well-founded subjective assignments of probability are precluded.

Kokshoorn and Luijsterburg [25] report a retrospective analysis of 74 requests for formal evaluative opinions on the probability of case findings given propositions at the activity level at the Netherlands Forensic Institute. Over half (57%) of the reports included one or more probability assignments that were based on expert opinion only, with relevant data not being available from peer-reviewed studies or other sources.

ENFSI also discuss the use of probabilities based solely on expert knowledge:

> Forensic practitioners often experience difficulty in assigning and justifying probabilities when the assignments are based on expert knowledge. However, likelihood ratios can be informed by subjective probabilities using expert knowledge. These probability assignments shall still be expressed by a number between 0 and 1 rather than by an undefined qualifier (such as frequent, rare, etc.). Such personal probability assignment is not arbitrary or speculative, but is based on a body of knowledge that should be available for auditing and disclosure. The forensic practitioner should not mislead the recipient of expert information as to the basis of the personal assignment, and the extent to which the assignment is supported by scientific research. Forensic practitioners should consider exploring the sensitivity of the likelihood ratio to different probabilities by examining the effect of assigning different probabilities according to their personal uncertainties.

We cannot describe a consensus here hence, we add Suggestion 4 (see ISFG recommendation 11; T&K section 10.3 discusses; FSR 7.2.10, OSAC, NIST dissent).

Suggestion 4: The community needs to determine whether a numerical value of the LR should be mandatory.

2.6.5 | Cross-examination or rebuttal

Cross-examination is not extensively discussed in the documents. The exceptions are FSR and NHF recommendation 6.3.

> FSR: 8.6.2 The expert should not, subject to the points below, give evidence on matters that were not addressed in their report(s). There may be cases where matters are raised in cross-examination, which are developed as a result of issues that have occurred during the trial, and which may need to be considered by the expert. Where there has been good case management, these should not be common.

> FSR: 8.6.6 If the expert is pressed to consider activity level questions that were not addressed in their report, they shall first consider whether or not they have the skills, knowledge, and information to do so. If they do not, they shall make the court aware that to proceed would be beyond the scope of their expertise. If they do, then they should ask for time to consider their interpretation. The court shall be made aware of the limitations, including the lack of peer review of this opinion and the fact that the expert is unable to consult sources of data such as publications. It is always preferable for activity level questions to have been addressed in the report if it is thought likely that these are the questions of relevance to the court. If necessary, the expert shall request sufficient time to return to their place of work to consider the issues, consult data, and seek peer review.

> ENSFI-BPM pg. 28: The expert shall not give opinions on matters that were not addressed in their report(s). There may be cases where matters are raised in cross-examination, which are developed as a result of issues that have occurred during the trial, and which may need to be considered by the expert. If an opinion is given or interpretations made fall outside the scope of the accreditation, or is outside the scope of his/her expertise, then this is stated as such by the scientist. Where there has been good case management, it is rare to give an opinion that is not available in the report.

> ENSFI-BPM at pg. 29: During court cross-examination, questions may arise that were not considered in the original statement, and new information is required to help the court. The scientist

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TABLE 3 Summary of consensus recommendations and suggestions.

1. Principles
Consensus recommendation 1: Principles of evidence evaluation: 1. Interpretation takes place in a framework of circumstances 2. We must evaluate our results given two mutually exclusive propositions representing the views of the two involved parties as understood 3. Scientists must give their opinion on the results and not on the propositions
2. Quality assurance/quality control
Consensus recommendation 2: Experts shall only answer questions that they have been shown to be qualified to answer and shall be able to recognize the progression from simple answerable questions to ones that require additional work, data, or training. Experts undertaking formal EGALP shall be trained to do so. They shall have demonstrated competency Consensus recommendation 3: All reports shall be peer reviewed
3. Case management
Consensus recommendation 4: Case pre-assessment should be conducted in cases when activity level propositions are set Consensus recommendation 5: The expert shall decide whether or not to proceed with EGALP based on the availability of expertise and resource, and on the potential to deliver value Consensus recommendation 6: The propositions should be based on case information and formulated according to published guidelines; for example, they shall not include explanations of how the DNA transferred, or results obtained Consensus recommendation 7: There shall be, in the report, a sentence saying that if the information (particularly that on which propositions are based) changes, then the interpretation will need to be reconsidered Consensus recommendation 8: To make clear the limitations of providing a report that is restricted sub-source level, the following caveat can be applied to statements: "This report does not provide any information on the mechanisms or actions that led to the deposition of the biological material concerned. It only provides help regarding its origin. Should there be any issue regarding the activities that led to this material, we should be contacted to provide a new report" Consensus recommendation 9: The defense should be given an opportunity to comment as early as realistic on the case information presented as understood, propositions, data sources, and any assumptions made
5. Assignment of the LR
Consensus recommendation 10: A likelihood ratio shall be assigned which compares the probability of the findings given the first proposition (and considering the available case information) to the probability of the findings given an alternate proposition (still considering the available case information). The propositions are ascribed to the prosecution and alternative (ideally defense) positions, respectively Consensus recommendation 11: Sources of data should ideally be a study that has conditions that match those of the case. Failing the availability of such data, the scientist may use secondary sources or expert opinion. Regardless of the source, it should be transparent within the report how the probabilities have been assigned Consensus recommendation 12: All results (e.g., presence and absence of DNA) shall be included in the observations being evaluated, or their omission (for example due to the irrelevance to the disputed activity) should be explained and justified Consensus recommendation 13: The case notes shall contain, at a minimum, the background case information, the propositions, all of the published material or sources of data, the logic, formulae, or Bayes Net used, the resulting *LR* or the results of the sensitivity analysis
6. Reporting
Consensus recommendation 14: The phrasing of the probabilistic statements in the report, notes, and testimony shall always be of the form: the probability of the results given that each (in turn) of the propositions is true. The phrasing of the conclusion shall pertain to the ratio of two probabilistic statements regarding the results Consensus recommendation 15: A report shall be issued wherever practical. The report shall include, at a minimum, the case information relied upon, the propositions used, and an assessment of the LR *in* a numerical form. A verbal qualifier is optional. The report shall include all relevant traceability information. The report should invite the defense to have input Consensus recommendation 16: Explicit reference to any scientific publications that give the data used, any unpublished data, any subjective adjustments to the data and the reason for this, and the logic used to assign the LR shall appear in the case notes and in the statement, perhaps in an appendix Consensus recommendation 17: Any unpublished data used shall be made available, if necessary under a non-disclosure agreement, for the use of anyone involved in the case. It is acceptable to anonymize DNA profiles. This may be impacted by local laws and ethics considerations
7. Testimony
Consensus recommendation 18: All DNA experts testifying about the value of DNA profiling comparison shall be equipped to give relevant warnings regarding evaluation given alleged activities Consensus recommendation 19: A prosecution witness shall not leave the court with the impression that there are a range of transfer mechanisms without at least warning the court that there are important and relevant issues

(Continues)

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TABLE 3 (Continued)

Suggestions
4. Setting the propositions
Suggestion 1: Propositions must be: 1. Relevant to the court 2. Consider all relevant routes for transfer, persistence and recovery of biological traces 3. Must be specific enough that assigning a probability of the evidence given the truth of them is possible 4. Come in (at least) pairs, (the "at least" is added by the authors of this document) 5. They must be exclusive and should be exhaustive within the context of the case 6. They should not include results or mechanisms 7. They should be concise
5. Assignment of the *LR*
Suggestion 2: If the analysis assumes that no contamination has occurred, then this shall be explicitly written into the notes and the report Suggestion 3: A sensitivity analysis may be performed. If performed it should be recorded in the notes and signaled in the report
6. Reporting
Suggestion 4: The community needs to determine whether a numerical value of the *LR* should be mandatory Suggestion 5: For prosecution witnesses called in rebuttal, a written report should be given when it is practical

> must state the limitations of the current report; then he/she can suggest that in order to answer the court query, further work is necessary which may be outlined.

We think that these FSR and ENSFI-BPM recommendations are reasonable but optimistic.

Guidance documents are silent on the subject of rebuttal evidence. There is often no legal requirement for written disclosure for this, and in the US, rebuttal evidence is generally exempted from discovery [26–29]. It may be impossible, due to timing, for a witness called by the prosecution to issue a report, properly peer reviewed and approved to rebut unforeseen testimony given at trial. In our own laboratories, this process could take weeks. However, we suggest that a written report should be issued wherever possible, and the analysts called by the party with the burden should advocate for the acceptance of a report by the counsel.

Suggestion 5: For prosecution witnesses called in rebuttal a written report should be given when it is practical.

2.7 | Testimony

2.7.1 | Explanations

The authors of this paper suggest that when faced with a decision to undertake a very difficult and subjective analysis, there is a very understandable tendency to not do it. However, it is important to balance the risk of undertaking this analysis against the risk of not undertaking it.

This has been clearly highlighted by, for example, the ISFG guidelines [2] as their recommendation 1.

ISFG Recommendation 1: *Providing a list of possible explanations for the results < snip > (is) not (relevant) in court, as it does not allow an assessment of the value of evidence.*

Unfortunately, this needs a little translation. It is a statement that we interpret to mean that bringing up various explanations at court (different from propositions, as now the results are known), say the transfer from social contact, is not relevant in court without some assessment of the relative probabilities of the evidence given this proposal and a proposal aligned with the prosecution allegation. Taken literally, and we agree, it is not appropriate to give a simple "yes" to questions such as "is transfer possible" by some proposed mechanism. The literal answer to this question is ALWAYS "yes" but this is exactly what the ISFG warn against. It is also possible that a defense witness brings up various scenarios as "possible" and potentially quotes any one of a number of studies that show this to be true. But again, this is exactly what the ISFG guidelines warn against. The likely effect is that various possibilities are placed before the fact finders with no weighting. What happens next in the decision process is unpredictable.

During cross-examination, the questions are often looking for answers that explain the results. These are termed explanations. While these explanations may give rise to the observed evidence, any answers given should address the probability of the evidence, and not that some explanation is possible. This is addressed by OSAC 4.4.4.2: *If a hypothetical question is posed in the form of "Is this possible?", the expert should communicate that the value of evidence is based on comparing probabilities of the results given competing propositions and not possibilities, and then give an answer in terms of relative probabilities.*

NHF has some suggested answers in their tab. 7.1 [6].

The Body Fluid Forum (BFF) is subgroup of the Association of Forensic Science Providers (AFSP). It consists of a representative from each of the Forensic Science Providers in the United Kingdom and Ireland [30]. It wrote:

> It is the opinion of the BFF that not evaluating DNA evidence in case work is potentially more dangerous and reckless than carrying out an evaluation based on

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limited datasets. The investigators and courts may be fixated on the probity of the DNA result in isolation and in doing so not think about other issues including the mode of transfer of the DNA evidence. We believe that it is our professional duty to communicate the complexities of the issues and the limitations, which are under consideration in any specific case and provide subjective guidance where possible, even with limited data sets. Shying away from this duty on the grounds that considerations regarding transfer of trace DNA is less known than source level DNA is not acceptable.

The authors of this paper observe that while there is safety for a witness in a simple "yes" to an "is secondary transfer possible" question, or in stating that they are unqualified to comment, there is no safety in this for the justice process as it creates a void that is most likely filled by the judge and jury fueled by whatever soundbites the legal advocates have offered. We suggest, as do the documents, that a more balanced answer is given maybe along the lines of OSAC 4.4.4.2.

Consensus recommendation 18: All DNA experts testifying about the value of DNA profiling comparison shall be equipped to give relevant warnings regarding evaluation given alleged activities.

Consensus recommendation 19: A prosecution witness shall not leave the court with the impression that there is a range of transfer mechanisms without at least warning the court that there are important and relevant issues.

For convenience, we have provided a summary of consensus recommendations and suggestions in Table 3.

3 | CONCLUSION

We believe that the emerging perception, at least in the US, is that EGALP is very difficult and fraught with pitfalls. Central to this perception are the setting of propositions with limited information, and the paucity of reliable data covering the required variables. These matters are aggravated by moderate divergences in recommendations from existing documents.

We have attempted discuss and compare existing guidance that results in 19 consensus recommendations. We have added a further three suggestions that arise, in part, from the documents but where a consensus is difficult to establish. In undertaking this comparison, we highlight:

1. A diversity of opinion regarding providing a numerical value for the LR,
2. A need to enhance guidance on cross-examination and rebuttal.

The standard implied by these recommendations is aspirational. We believe that avoiding questions that may help the fact finder determine how the DNA ended up where it was found also has strong negative potential. Hence, we suggest that it is necessary to empower acceptable partial conformance immediately.

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

Data sharing not applicable – no new data generated.

DISCLAIMER

Points of view in this document are those of the authors and do not necessarily represent the official position or policies of their organizations or funding agencies.

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